INTERNATIONAL MONETARY FUND

REPUBLIC OF MOZAMBIQUE

April 10, 2019

REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS[1]

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Republic of Mozambique Joint Bank-Fund Debt Sustainability Analysis	
Risk of External debt distress:	In debt distress
Overall risk of debt distress	In debt distress
Granularity in the risk rating	Sustainable
Application of judgment	No

Mozambique's debt remains in distress as debt restructuring discussions are ongoing with international private creditors. Accounting for the authorities' debt restructuring strategy and considering that, to a large extent, future borrowing and government guarantees reflect state participation in the sizable liquified natural gas (LNG) development, debt is deemed sustainable in a forward-looking sense. The present value (PV) of external public and publicly guaranteed (PPG) debt relative to GDP is projected to surpass the prudent thresholds for a few years but remains on a gradual declining path; while the PV of PPG external debt relative to exports remain close to the prudent threshold. Both debt service ratios (to exports and revenue) are projected to remain below their respective thresholds during the projection period.

[1] Mozambique's Composite Indicator score is 2.64, indicating that the country's debt-carrying capacity is weak. The analysis presented in this document is based on the new debt sustainability framework (DSF) for low-income countries (http://www.imf.org/en/Publications/Policy-Papers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf).

PUBLIC DEBT COVERAGE

1. Compared to the last DSA,[2] the coverage of public and publicly guaranteed debt remains unchanged. The scope of Mozambique's public debt covers data on external and domestic obligations of the central and local governments. The authorities also provided debt data on state-owned enterprises (SOEs) and SOE debt guarantees.[3] Domestic debt is denominated in local currency and, for the purposes of the analysis, domestic debt is assessed by currency and not by residence as data are unavailable to capture the residency of creditors. The main source of uncertainty is the level of domestic payments arrears,[4] and delayed processing of value added tax (VAT) refunds.[5]

Text Table 1. Mozambique Public Debt Coverage Under the Baseline Scenario

	Subsectors of the public sector	Sub-Sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

BACKGROUND

A. Recent Economic and Debt Developments

2. Mozambique's economy has been improving gradually. Real GDP growth reached 3¾ percent in 2017 supported by a stronger recovery in agriculture and higher mining production. However, real growth decelerated to 3¼ percent because of deceleration in mining growth but became broader based. Tight monetary policy and lower food price increases brought annual inflation down to low levels (3½ percent at end-2018). The non-megaproject current account deficit has been narrowing, and the central bank has rebuilt its international reserves to a comfortable level (6¼ months of next year's non-megaprojects imports at end-2018).

3. The stock of public debt in 2017 and its estimate for 2018 are lower compared to the last DSA. A lower fiscal deficit and an easing of the monetary policy stance helped to reduce government borrowing needs (Text Table 2). The fiscal effort included cuts in public investment as well as the

[2] IMF Country Report 18/32, March 7, 2018.

[3] This includes Mozambique's National Hydrocarbons Company (ENH) borrowing to finance its equity participation in the LNG megaprojects, along with any sovereign guarantees issued to cover ENH's share in the LNG megaprojects' financing package.

[4] As of end-December 2017, the stock of domestic payments arrears was around 19 billion meticais (2½ percent of GDP), of which 13.4 billion meticais (1½ percent of GDP) had been validated.

[5] As of end-September 2018, outstanding VAT refunds stood at 22.4 billion meticais (2½ percent of GDP).

elimination of subsidies on fuels and wheat, the adoption of an automatic fuel price adjustment mechanism, and increases in electricity and public transportation tariffs. In addition, the authorities scaled back considerably, vis-à-vis earlier expectations, the contracting of external debt for large projects, including on-lending to state owned enterprises.

Text Table 2. Mozambique: Debt, Public and Publicly Guaranteed

(Percent of GDP)

	2014	2015	2016	2017 Prel.	2018 Est.
Public sector debt [1]	**67.2**	**94.2**	**137.6**	**110.4**	**110.5**
Public sector external debt (incl. guarantees)	**60.7**	**82.5**	**112.9**	**90.4**	**91.2**
A. Bank of Mozambique-IMF	1.1	1.9	2.2	1.4	1.1
B. General Government	49.6	68.5	98.0	76.1	74.6
Multilateral creditors, excl IMF	20.8	28.2	39.4	30.8	30.0
Bilateral creditors [1]	20.8	30.9	42.7	33.8	32.4
Paris Club	3.3	5.1	7.8	7.5	6.6
Banks	3.2	3.3	7.5	5.4	5.1
EMATUM/Mozam Eurobond	3.2	3.3	7.5	5.4	5.1
Other public sector: *ENH (LNG project)*	4.8	6.1	8.4	6.2	7.0
C. Government guaranteed external debt	9.9	12.1	10.8	7.6	7.2
EMATUM	2.2	2.7	0.0	0.0	0.0
Proindicus	3.9	4.8	6.2	4.4	4.2
MAM	3.4	4.2	4.2	3.0	2.8
Other guarantees	0.4	0.4	0.5	0.3	0.2
D. External arrears	0.0	0.0	1.9	5.2	8.3
Public sector domestic debt (incl. guarantees)	**6.5**	**11.7**	**24.7**	**20.0**	**19.2**

[1]This reflects preliminary numbers for the stock of debt at end-2018 shared with IMF Staff in March 2019. Domestic debt includes domestic debt of a selected number of SOEs for 2016-18, for which data was available. The selection criteria are set in the Joint IMF-Bank Guidance Note on Low-Income Countries Debt Sustainability Framework. The figures includes ENH borrowing from external partners for its equity participation in Area 4 Coral South and Mamba LNG projects, Area 1 Golfinho Atum project, and the state guarantee for ENH debt for its participation in the Area 1 Golfinho Atum LNG project.

4. The authorities are in good-faith discussions with private creditors to restructure Mozambique's debt. An agreement in principle with a group of Eurobond holders was reached last November to provide debt service relief over the next five years (before production and exports from the LNG megaprojects are expected to start) and extend principal repayments by ten years. Renegotiation discussions on the government-guaranteed loan contracted by MAM, another SOE, are nearly finalized. The new restructuring terms would provide significant debt service relief and staff assess that there is a good prospect of these discussions concluding as envisaged. At the same time, Mozambique's Attorney-General has filed a lawsuit in the U.K. to nullify the government guarantee to the loan contracted by Proindicus with Credit Suisse, in light of the indictments in the U.S. and in Mozambique that allege criminal behavior in the contracting of this loan.[6] The authorities intend that any agreements with creditors will protect the interests of Mozambique if future developments call into question the legality of the contracting of the loans or the issuance of the government guarantees.

[6] The authorities have conveyed to staff that the government does not intend to support Proindicus, which will follow the normal course of commercial bankruptcy without backing. Therefore, the loan has been removed from the DSA baseline. Further, given the authorities' representation that the validity of the guarantee is in dispute, the guarantee has also been removed from the DSA baseline and is instead treated as a contingent liability in accordance to IMF policy (Text Table 3).

Text Table 3. Mozambique: Contingent Liability

1	The country's coverage of public debt	The central, state, and local governments plus social security, central bank, government-guaranteed debt, non-guaranteed SOE debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP		
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	4.55	SOE sovereign guarantee in dispute
4	PPP	35 percent of PPP stock	3.21	
5	Financial market (the default value of 5 percent of GDP is the minimum v	5 percent of GDP	5.00	
	Total (2+3+4+5) (in percent of GDP)		12.76	

5. Mozambique continues to accumulate external debt service arrears on the defaulted loans. The overall stock of external arrears on public and publicly guaranteed external debt service is estimated to have reached US$1,178 million (8.3 percent of GDP) at end-2018 (Text Table 4).[7] This also includes arrears under bilateral discussion with five official creditors amounting to US$181 million (Libya, Iraq, Angola, Bulgaria and Poland),[8] and arrears amounting to US$26 million with Brazil on the US$125 million government-guaranteed borrowing undertaken by the state-owned airport company for which the guarantee has been called.[9, 10] The government is servicing all the remaining multilateral and bilateral external debt obligations.

Text Table 4. Mozambique: Evolution of the Stock of External Arrears

(Millions of U.S. dollars)[1]

	End-2017 Est.	End-2018 Est.
Commercial debt	**592.86**	**970.61**
Mozam/ EMATUM	97.90	174.18
MAM	343.06	499.12
Proindicus	151.90	297.31
Bilateral debt	116.80	207.33
Paris Club: Brazil	22.93	37.65
Non-Paris Club [2]	93.87	169.68
Total	**709.66**	**1177.94**

[1] Staff estimates based on information received by the authorities on their debt strategy. Contractual penalty fees or rates have not
[2] Data reported by the authorities for Libya, Iraq, Angola, Poland and Bulgaria on the reconciled debt.

B. Macroeconomic Forecasts

6. The macroeconomic assumptions underlying the baseline scenario for the next five years have improved relative to the last DSA and accounts for adverse economic effects of Cyclone Idai. Overall, the economy has been expanding faster than earlier projections. Preliminary estimates suggest that real GDP growth in 2019 could decline to a range of 1.8 percent to 2.8 percent—down from a pre-cyclone projection of 3.8 percent in 2019—owing to significant losses to agricultural production and disruptions to transport, communications and services. Growth is projected to rebound strongly in 2020 in

[7] For the Eurobond, the accumulated arrears and the principal of US$726 million equals the principal of US$900 million mentioned in the November 2018 agreement in principle.

[8] The arrears with Libya, Iraq, Angola, Bulgaria and Poland are not new arrears. The authorities have reconciled the debt with these non-Paris Club creditors dating back the HIPC Initiative in 2006, 2011 and 2015, respectively. The HIPC Initiative resulted in a representative Paris Club Agreed Minute, and the authorities are pursuing best efforts to reach agreement on a restructuring with the remaining non-Paris Club creditors. While the negotiations are ongoing, the debt payment obligations incurred from the date of reconciliation until end-2017 have been included in the stock of arrears in line with the terms of the reconciliation registered in authorities' external debt database.

[9] Any contractual penalty fees or rates on missed payments have not been included in the estimate for the stock of arrears reflected in the baseline.

[10] The two loans from the Brazilian Development Bank (BNDES) to the state-owned airports company (ADM) were also guaranteed to BNDES by the Brazilian Export Guarantee Fund (this guarantee has also been called).

response to a recovery in agricultural production and reconstruction efforts, and to reach 4 percent annually in 2021-22 (Text Table 5). The overall fiscal deficit is projected to increase in 2019 because of the budgetary effects of Cyclone Idai and improve thereafter in line with the fiscal effort to eliminate the fiscal primary deficit after grants by 2022 and, subsequently, as fiscal revenue from LNG production materializes. The external current account deficit is projected to increase to 76 percent of GDP by 2022, reflecting a surge in imports of goods and services mainly related to the construction of processing facilities for the LNG megaprojects. Approximately one-third of all imports related to such projects will be financed by (frontloaded) FDI, with the remainder financed by private debt. With the start of LNG production in 2023, growth will pick up significantly. However, relative to the last DSA, long-term projections are more conservative, as the underlying scale of LNG production has been revised downward (Box 1).

Text Table 5. Evolution of Selected Macroeconomic Indicators Between DSA Updates[1]

	2016	2017	2018	2019	2020	2021	2022	2023	2024	2028	2037	2038
	Prel.	Est.	Est.	Projections								
Real GDP growth (percent)[2]												
Previous DSA: 2017 A4	3.8	2.9	3.0	2.5	2.4	2.3	2.2	9.9	31.6	6.5	1.9	...
Current DSA	3.8	3.7	3.3	1.8	6.0	4.0	4.0	9.2	11.5	2.8	2.6	2.7
Nominal GDP (US$ billion)												
Previous DSA: 2017 A4	11.3	12.7	14.3	15.0	15.7	16.4	17.1	19.1	25.6	65.3	95.1	...
Current DSA	10.9	12.6	14.4	15.1	16.7	17.9	19.0	21.3	24.2	41.3	69.5	72.8
Overall fiscal balance (percent of GDP)												
Previous DSA: 2017 A4	-7.6	-5.5	-7.5	-10.8	-9.9	-9.6	-10.6	-9.9	-6.8	-1.8	0.4	...
Current DSA	-7.3	-3.4	-5.5	-6.5	-4.8	-3.7	-2.6	-1.9	-0.1	1.8	14.4	14.0
Current account deficit (percent of GDP)												
Previous DSA: 2017 A4	39.2	16.1	16.9	44.6	66.6	91.0	116.4	117.7	78.0	-26.2	-35.9	...
Current DSA	39.0	20.0	30.4	58.0	66.7	62.9	75.6	61.6	39.3	-1.7	-4.9	-5.8
FDI (percent of GDP)												
Previous DSA: 2017 A4	27.4	12.1	12.4	34.5	53.4	75.0	19.8	3.8	2.3	0.5	0.8	...
Current DSA	28.4	18.2	18.7	18.5	21.2	20.1	23.9	17.1	13.5	3.2	5.5	6.0

Sources: Mozambican authorities, and IMF staff estimates and projections.
[1] Lower forcasts reflect a downward revision in projected number of trains.
[2] Real GDP jumps in 2023 due to the expected coming onstream of the LNG production.

7. The risks to the medium-term macroeconomic outlook are broadly balanced. Downside risks include a deterioration in security conditions in the North; backtracking of peace negotiations; loosening of expenditures control in the run-up to the October elections; delays in megaprojects; and future extreme climate events. New oil and gas discoveries; a quicker resolution of ongoing debt renegotiations; and closer reengagement with donors would constitute upside risks.

8. According to the debt sustainability framework's realism tools, the baseline projections are reasonable (Figures 3 and 4). The growth projections are relatively lower than suggested by the alternative fiscal-growth multiplier analysis due to the adverse growth effects related to tropical cyclone Idai. The baseline fiscal adjustment in the primary balance seems credible as it does not fall in the upper quartile of the distribution of past adjustments. The contribution of public investment to growth seems reasonable in comparison to historical contributions and the gradual economic recovery following the 2016 economic downturn.

Box 1. Key Assumptions, 2018–38

Real GDP growth is to decline in 2019 in line with the adverse economic effects related to Cyclone Idai. Thereafter, growth is assumed to rebound sharply in 2020 and remain around 4 percent in 2021-22. A sharp increase in growth in 2023 reflects the onset of production from the first LNG plant (train) and related exports in that year. Other economic activity (non-LNG) is conservatively assumed to continue to grow at a steady rate of 4 percent per year over the long-term.

LNG sector. Start of LNG production from Area 4 is assumed in 2023 and from Area 1 in 2024. However, two assumptions were revised compared to the last DSA: (i) five LNG production trains are assumed to begin production over 2023-30, instead of 16 trains previously; and (ii) the development cost for these trains were revised upward in line with new market conditions. A lower number of trains entails a lower contribution to GDP, fiscal revenue, exports, FDI, and private external financing. Higher development costs imply a higher debt that needs to be incurred, including by ENH to finance its equity participation share in the LNG megaprojects.

Consumer price inflation is projected to remain—after a temporary spike in 2019-20 given the supply-side effects of Cyclone Idai on consumer prices—within the authorities' projection range of 5–6 percent per year.

Fiscal. The primary fiscal deficit after grants is projected to temporarily deteriorate in 2019 given the effects of Cyclone Idai and improve thereafter in line with (i) the authorities' fiscal adjustment efforts (to around zero percent of GDP by 2022), and (ii) the onset of LNG production in 2023. Tax and non-tax revenue as well as primary expenditure are projected to grow in line with nominal non-LNG GDP over the long-term. As revenue from LNG production flows in by 2023, the primary fiscal balance would improve and turn into surplus reaching 13¼ percent of GDP by 2038. Underlying the path of improvement of the primary fiscal balance is the assumption that (i) the recovery of LNG development costs would take place over the initial four-to-six years of production, after which LNG fiscal revenue would pick up significantly and account for almost half of total fiscal revenue, and (ii) all LNG fiscal revenue would be saved.[1]

External sector. The current account deficit is projected to peak at over 76 percent of GDP by 2022 driven by LNG megaproject imports. This deficit would be primarily financed through FDI and private external borrowing. The very large size of LNG exports would turn current account deficits into surpluses by 2027.

Debt restructuring. The baseline scenario includes the effect on debt and debt service payments of the agreement in principle reached with a group of bondholders on restructuring Mozambique's Eurobond, and the authorities' debt strategy regarding the government guaranteed loans taken up by Proindicus and MAM with Credit Suisse and VTB, respectively as noted in paragraph 4.

1/ There is currently no political decision on the share of LNG related fiscal revenue that would be saved for future generations and considering capacity constraints. While the assumption that all this revenue would be saved seems extreme, the impact of this assumption on the evolution of debt indicators is mitigated by the assumption that public debt would not be pre-paid and the fact that, by saving these resources, non-LNG growth would remain unchanged only at 4 percent a year over the long-term.

COUNTRY CLASSIFICATION

9. Mozambique's debt carrying capacity is assessed as weak, compared to medium in the last DSA (Text Table 6).[11] In the previous DSA, Mozambique's debt carrying capacity was measured solely by the World Bank's Country Policy and Institutional Capacity (CPIA) index. A feature of the new DSF is a composite indicator (CI) that draws on the CPIA and other country-specific factors, including the ratio of international reserves to total imports. For the case of Mozambique, the reason for the change in the

[11] This downward revision of the debt carrying capacity has been informed by two weak signals (Fall 2018 and Spring 2019 WEOs).

debt carrying capacity assessment is the huge scale of LNG megaprojects under development (these projects amount to four times Mozambique's GDP) and their related imports.[12] Accordingly, the relevant indicative threshold for the weak category are 30 percent for the PV of debt-to-GDP ratio, 140 percent for the PV of debt-to-exports ratio, 10 percent for the debt service-to-exports ratio, and 14 percent for the debt service-to-revenue ratio (Text Table 7). These thresholds are applicable to the public and publicly guaranteed external debt. The benchmark for the PV of total public debt under medium debt carrying capacity is 35 percent.

Text Table 6. Mozambique: Calculation of the Composite Indicator

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.293	1.27	48%
Real growth rate (in percent)	2.719	4.818	0.13	5%
Import coverage of reserves (in percent)	4.052	24.434	0.99	38%
Import coverage of reserves^2 (in percent)	-3.990	5.970	-0.24	-9%
Remittances (in percent)	2.022	0.343	0.01	0%
World economic growth (in percent)	13.520	3.579	0.48	18%
CI Score			2.640	100%
CI rating			Weak	

Text Table 7. Public and Publicly Guaranteed (PPG) External Debt Thresholds and Total Public Debt Benchmarks.

EXTERNAL debt burden thresholds	Weak	Medium	Strong
PV of debt in % of			
Exports	140	180	240
GDP	30	40	55
Debt service in % of			
Exports	10	15	21
Revenue	14	18	23

TOTAL public debt benchmark	Weak	Medium	Strong
PV of total public debt in percent of GDP	35	55	70

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

10. Under the baseline scenario, the solvency external debt indicators breach the policy relevant thresholds in the near and medium term (Table 1, and Text Table 8).[13]

[12] Mozambique's debt carrying capacity assessment would have remained medium if only non-megaproject imports were considered. Megaproject imports are fully financed through special investment vehicles outside the country and included in the balance of payments statistics for completeness of the information, with no potential bearing on international reserves. This would argue for the exercise of judgment in assessing sustainability.

[13] As in the last DSA, the historical scenario has been excluded from Figure 1. Such a scenario shows unrealistically fast declines in the public debt-to-GDP ratio, as the non-interest current account deficit at the historical average is much lower than projected because of LNG investments. Meanwhile, private debt accumulation is assumed to remain unchanged compared to the baseline.

- **The PV of external public debt in terms of GDP** is projected to remain above the prudent threshold over the medium term. This is partially driven by ENH borrowing to finance its equity participation in the LNG megaprojects, and the issuance of a sovereign guarantee to ENH to cover its share in the borrowing package for Area 1 Anadarko.[14] Equity participation borrowing by ENH includes additional borrowing for a third LNG project (Area 1 ExxonMobil) for which information was not accounted for in the last DSA. The ratios would drop below the prudent threshold by 2030, as the sovereign guarantee contractually lapses one-year after the start of LNG production in 2025. However, excluding ENH-related borrowing and sovereign guarantee, the ratios would follow a gradually declining path and fall under the prudent threshold earlier (by 2028).

- With coal exports projected to recover from 2019, **the PV of external public debt in terms of exports** would remain slightly above the 140 percent prudent threshold—and would improve rapidly thereafter as LNG exports pick up over the long term. Excluding ENH, the ratios would drop significantly below the prudent threshold and decline gradually over the medium term.

- **External public debt service in terms of revenue** would breach the prudent threshold in the near term and would remain around the prudent threshold over the medium term. The uptick around 2025 is a result of ENH starting to service its debt as LNG production is expected to start.

- **External public debt service in terms of exports** would drop below the prudent threshold in 2019 and would remain below it during the projection period.

Text Table 8. Mozambique: PPG External Debt
(Percent of GDP)

	2016	2017	2018	2019	2020	2021	2022	2023	2024
		Prel.	Est.			Proj.			
Public sector debt (Percent of GDP)									
Nominal stock of total debt	137.6	110.4	110.5	117.0	111.4	109.0	109.2	102.1	89.8
of which: external	112.9	90.4	91.2	97.9	96.3	96.9	98.6	94.5	85.7
Nominal stock of total debt, excl. ENH's debt	129.3	104.2	103.4	103.8	93.3	87.1	83.1	75.1	65.3
of which: external debt, excl. ENH's debt	104.5	84.2	84.2	84.7	78.2	75.0	72.6	67.5	61.2

Sources: Mozambican authorities, and IMF staff estimates and projections.

11. External public debt and public debt service ratios are most sensitive to exchange rate and export shocks (Figure 1 and Table 3). The stress tests illustrate that a nominal export growth (in U.S. dollars) set to its historical average minus one standard deviation, or the baseline projection minus one standard deviation, would increase the PV of external public debt-to-GDP ratio by 23 percentage points (to 92 percent) in 2020. Similarly, the PV of external public debt-to-

[14] In the baseline, the guarantee is phased-in in line with the expected flow of disbursements as the LNG megaproject led by Anadarko is implemented, and that contractually the guarantee does not give rise to debt service payments and it ceases to exist in the first year after LNG production starts.

export ratio would almost double to 330 percent in 2020 (compared to 171 percent under the baseline). The export shock has a similar effect on the two-debt service ratio. Vulnerability to exchange rate movements is an expected outcome for a commodity exporter like Mozambique with a relatively low export-to-exchange-rate elasticity and a relatively high share of external public debt in the total. External debt service indicators are also sensitive to the assumptions around ENH's future debt service profile.

B. Public Sector Debt Sustainability Analysis

12. The PV of debt-to-GDP ratio is projected to surpass the 35 percent of GDP threshold in the near and medium term (Table 2 and Figure 2). Total public and publicly guaranteed debt amounted to about 110 percent of GDP as of end-2018, with external debt accounting for about four-fifths of the total. The evolution of public debt is driven mainly by the same factors influencing external debt, including those related to ENH's share in the LNG megaprojects.

13. The standardized sensitivity analysis shows public debt indicators are highly sensitive to depreciation, as well as changes in current transfers and FDI (Table 4). A one-time 30 percent nominal depreciation of the metical would lead the PV of debt-to-GDP ratio to reach 105 percent (compared to 91 percent under the baseline scenario) by 2019. A shock to current transfers-to-GDP and FDI-to-GDP ratios (set to their historical averages minus one standard deviation in 2019 and 2020) would lead to a similar increase in the PV of debt-to-GDP ratio.

RISK RATING AND VULNERABILITIES

14. The DSA suggests that Mozambique's external public debt is in distress, but sustainable in a forward-looking sense. The debt service relief to be provided by the envisaged debt restructuring will suppress external debt service to sustainable levels and would provide the authorities with more time to implement their fiscal consolidation strategy. Together with the coming on-stream of the LNG projects, this is expected to put the public debt on a firmly downward trajectory over the medium-term. Moreover, as noted above, debt carrying capacity is likely to be stronger than signaled by the CI rating because international reserves do not need to cover imports related to megaprojects (which are fully funded through special investment vehicles outside the country). The sustainability assessment is anchored in the authorities' strong commitment to implement fiscal consolidation and a prudent borrowing strategy. The standardized stress tests indicate that an export shock and depreciation of the metical would lead to a deterioration of the debt position, underscoring the importance of diversifying the economy and broadening its export base.

15. Debt management and oversight needs to be strengthened further. While the authorities have already taken important steps, including to improve the transparency of the process of considering and granting sovereign guarantees, the MOF debt unit needs to be strengthened with respect to both its capacity and clout to exercise effective oversight over the entire public debt portfolio and to implement stronger safeguards. In addition, building on the new SOE Law, oversight over the entire SOE sector needs to be substantially strengthened.

AUTHORITIES' VIEWS

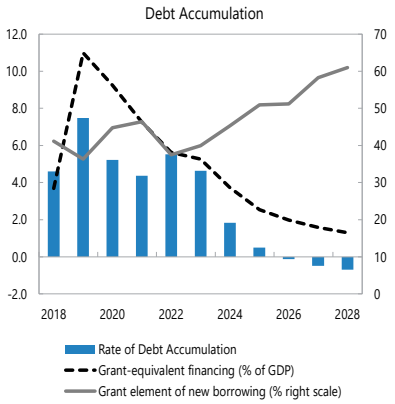
16. The authorities broadly agree with the analysis and recommendations above. They noted that their debt strategy is being implemented with the aim to bring Mozambique's debt risk to moderate levels over the medium term. This would be achieved through debt restructuring that would bring significant debt relief as well as medium-term fiscal consolidation. The authorities agreed with staff's recommendation to strengthen governance, transparency and accountability along the lines of the ongoing governance and corruption challenges diagnostic. They also acknowledged the importance of developing a framework for managing efficiently and transparently natural resource wealth and were appreciative of IMF support in this area, including the recent high-level conference on managing resource rich economies co-organized by the Fund and Bank of Mozambique.

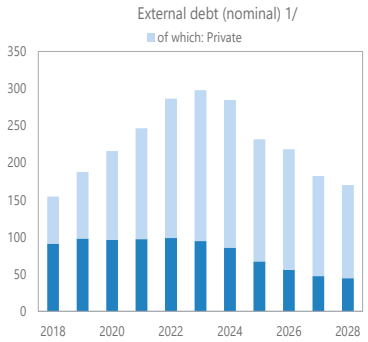
Table 1. Mozambique: External Debt Sustainability Framework Baseline Scenario, 2015-38

(In percent of GDP; unless otherwise indicated)

	Actual			Projections									Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2028	2038	Historical	Projections
External debt (nominal) 1/	**131.1**	**178.5**	**149.3**	**154.6**	**187.5**	**215.8**	**246.3**	**286.3**	**297.9**	**284.6**	**170.1**	**21.0**	**102.6**	**225.0**
of which: public and publicly guaranteed (PPG)	**82.5**	**112.9**	**90.4**	**91.2**	**97.9**	**96.3**	**96.9**	**98.6**	**94.5**	**85.7**	**44.4**	**13.9**	**56.8**	**79.6**
Change in external debt	32.6	47.4	-29.1	5.2	33.0	28.2	30.6	39.9	11.7	-13.4	-12.0	-5.7		
Identified net debt-creating flows	**28.0**	**57.5**	**-22.2**	**7.4**	**37.7**	**37.6**	**35.2**	**41.8**	**20.8**	**-4.8**	**-9.6**	**-12.4**	**5.4**	**6.0**
Non-interest current account deficit	**38.7**	**35.6**	**17.3**	**28.3**	**54.2**	**64.1**	**58.9**	**70.8**	**57.7**	**35.2**	**-7.5**	**-7.5**	**26.9**	**32.4**
Deficit in balance of goods and services	43.7	37.7	22.4	30.6	60.1	68.1	61.4	72.5	59.1	36.6	-15.5	-20.5	**33.1**	**33.1**
Exports	27.9	34.6	42.9	41.5	40.3	39.9	39.5	38.1	41.2	45.7	61.0	59.5		
Imports	71.7	72.3	65.4	72.1	100.3	108.0	100.9	110.6	100.3	82.3	45.5	39.0		
Net current transfers (negative = inflow)	-5.4	-2.2	-5.0	-2.3	-6.2	-4.3	-2.9	-1.9	-1.6	-1.5	-0.4	-0.2	**-6.7**	**-2.8**
of which: official	-3.6	-1.4	-3.7	-0.7	-4.6	-2.9	-1.6	-0.5	-0.4	-0.3	-0.1	0.0		
Other current account flows (negative = net inflow)	0.4	0.1	-0.1	0.1	0.3	0.4	0.4	0.2	0.2	0.0	8.4	13.2	**0.5**	**2.1**
Net FDI (negative = inflow)	**-26.1**	**-28.4**	**-18.2**	**-18.7**	**-18.5**	**-21.2**	**-20.1**	**-23.9**	**-17.1**	**-13.5**	**-3.2**	**-6.0**	**-22.7**	**-13.3**
Endogenous debt dynamics 2/	**15.4**	**50.3**	**-21.2**	**-2.2**	**2.0**	**-5.4**	**-3.6**	**-5.1**	**-19.7**	**-26.4**	**1.0**	**1.1**		
Contribution from nominal interest rate	1.6	3.4	2.7	2.0	4.6	4.8	4.5	4.1	3.9	3.6	5.8	1.8		
Contribution from real GDP growth	-7.4	-6.7	-5.8	-4.3	-2.7	-10.2	-8.1	-9.3	-23.7	-30.0	-4.8	-0.7		
Contribution from price and exchange rate changes	21.2	53.6	-18.2	...	...	...	...	...	...	...	...	...		
Residual 3/	**4.6**	**-10.1**	**-6.9**	**-2.2**	**-4.7**	**-9.3**	**-4.6**	**-1.8**	**-9.2**	**-8.6**	**-2.4**	**6.7**	**1.3**	**-4.1**
of which: exceptional financing	0.0	-1.7	-3.0	-2.2	-1.5	-0.5	-0.5	-0.4	-0.4	-0.2	-0.1	0.0		
Sustainability indicators														
PV of PPG external debt-to-GDP ratio	**...**	**...**	**71.4**	**66.4**	**70.5**	**68.5**	**68.0**	**69.1**	**65.9**	**59.5**	**34.0**	**11.6**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**166.3**	**160.1**	**175.1**	**171.5**	**171.9**	**181.2**	**159.9**	**130.3**	**55.8**	**19.5**		
PPG debt service-to-exports ratio	**8.3**	**15.8**	**10.9**	**11.7**	**9.8**	**9.4**	**9.4**	**9.3**	**9.2**	**6.9**	**4.9**	**2.0**		
PPG debt service-to-revenue ratio	**9.3**	**22.6**	**17.9**	**20.1**	**16.4**	**14.9**	**14.3**	**13.6**	**14.3**	**12.4**	**14.1**	**3.6**		
Gross external financing need (Million of U.S. dollars)	2601.8	1796.6	709.0	2300.9	6757.7	8758.9	8565.5	10570.2	10438.0	7037.8	2110.7	-3875.4		
Key macroeconomic assumptions														
Real GDP growth (in percent)	6.6	3.8	3.7	3.3	1.8	6.0	4.0	4.0	9.2	11.5	2.8	2.7	**6.3**	**8.1**
GDP deflator in US dollar terms (change in percent)	-17.7	-29.0	11.3	10.8	3.0	4.3	3.0	2.4	2.3	2.2	2.0	2.0	**-2.1**	**3.3**
Effective interest rate (percent) 4/	1.4	1.9	1.7	1.5	3.2	2.9	2.2	1.8	1.5	1.4	3.3	7.0	**1.9**	**2.2**
Growth of exports of G&S (US dollar terms, in percent)	-10.9	-8.9	43.4	10.5	1.8	9.6	6.1	2.6	20.9	26.1	3.1	4.4	**8.3**	**16.0**
Growth of imports of G&S (US dollar terms, in percent)	-13.4	-25.8	4.5	26.2	45.9	19.1	0.1	16.6	1.4	-6.6	-1.1	0.5	**11.7**	**8.8**
Grant element of new public sector borrowing (in percent)	...	...	...	41.1	36.3	44.8	46.4	37.5	39.9	45.3	61.0	61.0	**...**	**46.6**
Government revenues (excluding grants, in percent of GDP)	25.0	24.1	26.2	24.2	24.0	25.1	25.8	26.0	26.5	25.4	21.3	32.9	**21.8**	**24.2**
Aid flows (in Million of US dollars) 5/	1283.9	671.1	1450.0	743.0	1881.5	1870.1	1636.1	1372.0	1563.7	1274.5	674.7	361.3		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	3.7	11.0	9.2	7.3	5.6	5.3	3.7	1.3	0.4	**...**	**4.8**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	57.2	58.6	63.1	59.8	48.6	51.0	61.1	79.4	77.7	**...**	**62.9**
Nominal GDP (Million of US dollars)	14,798	10,897	12,586	14,396	15,093	16,695	17,879	19,032	21,277	24,228	41,313	72,794		
Nominal dollar GDP growth	-12.3	-26.4	15.5	14.4	4.8	10.6	7.1	6.5	11.8	13.9	4.8	4.7	**4.1**	**11.6**
Memorandum items:														
PV of external debt 7/	...	...	130.4	129.8	160.1	187.9	217.4	256.7	269.3	258.4	159.7	18.7		
In percent of exports	...	...	303.6	312.8	397.5	470.6	549.7	673.6	653.2	565.8	262.1	31.4		
Total external debt service-to-exports ratio	17.8	26.8	15.4	15.3	22.6	23.8	23.0	22.6	20.6	16.2	25.8	13.8		
PV of PPG external debt (in Million of US dollars)			8986.0	9564.4	10641.2	11428.8	12156.6	13143.8	14025.2	14414.4	14059.4	8459.8		
(PVt-PVt-1)/GDPt-1 (in percent)				4.6	7.5	5.2	4.4	5.5	4.6	1.8	-0.7	-0.5		
Non-interest current account deficit that stabilizes debt ratio	6.1	-11.8	46.4	23.1	21.2	35.9	28.3	30.9	46.0	48.6	4.6	-1.8		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

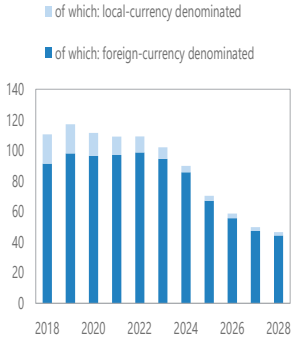
Table 2. Mozambique: Public Debt Sustainability Framework, Baseline Scenario, 2015-38

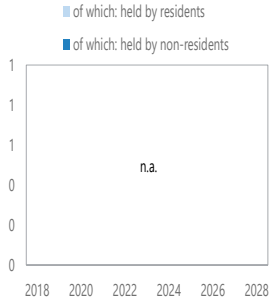
(In percent of GDP; unless otherwise indicated)

	Actual			Projections									Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2028	2038	Historical	Projections
Public sector debt 1/	**94.2**	**137.6**	**110.4**	**110.5**	**117.0**	**111.4**	**109.0**	**109.2**	**102.1**	**89.8**	**46.5**	**21.8**	**66.2**	**88.6**
of which: external debt	**82.5**	**112.9**	**90.4**	**91.2**	**97.9**	**96.3**	**96.9**	**98.6**	**94.5**	**85.7**	**44.4**	**13.9**	**56.8**	**79.6**
				19.2	19.1	15.1	12.1	10.5	7.5	4.1	2.1	7.9		
Change in public sector debt	27.0	43.4	-27.3	0.1	6.6	-5.6	-2.4	0.2	-7.1	-12.3	-3.3	-0.4		
Identified debt-creating flows	**21.9**	**35.5**	**-29.7**	**3.4**	**5.7**	**-3.5**	**-1.2**	**-1.8**	**-7.9**	**-10.8**	**-4.0**	**-14.8**	**5.0**	**-5.4**
Primary deficit	**7.2**	**6.0**	**3.1**	**5.2**	**6.5**	**4.8**	**3.7**	**2.6**	**1.5**	**-0.2**	**-3.1**	**-14.6**	**4.9**	**1.0**
Revenue and grants	28.1	26.2	28.3	26.0	30.6	30.0	28.9	28.0	28.5	27.1	22.1	33.1	**27.2**	**26.6**
of which: grants	3.0	2.1	2.0	1.8	6.6	4.9	3.0	2.0	1.9	1.8	0.8	0.2		
Primary (noninterest) expenditure	35.2	32.2	31.4	31.2	37.1	34.7	32.6	30.6	30.0	27.0	19.1	18.5	**32.1**	**27.6**
Automatic debt dynamics	**14.7**	**29.5**	**-32.8**	**-1.8**	**-0.8**	**-8.3**	**-5.0**	**-4.4**	**-9.4**	**-10.6**	**-1.0**	**-0.2**		
Contribution from interest rate/growth differential	-3.7	-2.9	-5.4	-2.4	-0.8	-6.1	-3.9	-4.0	-9.1	-10.5	-1.0	-0.2		
of which: contribution from average real interest rate	0.5	0.5	-0.4	1.0	1.1	0.6	0.4	0.2	0.1	0.0	0.4	0.4		
of which: contribution from real GDP growth	-4.2	-3.4	-5.0	-3.5	-2.0	-6.7	-4.3	-4.2	-9.2	-10.5	-1.3	-0.6		
Contribution from real exchange rate depreciation	18.4	32.4	-27.5	...	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**5.1**	**7.9**	**2.4**	**-2.7**	**0.9**	**-4.3**	**-2.3**	**1.5**	**0.5**	**-1.6**	**0.8**	**14.4**	**2.4**	**-0.8**
Sustainability indicators														
PV of public debt-to-GDP ratio 2/	**...**	**...**	**86.2**	**86.4**	**90.9**	**84.7**	**81.1**	**80.6**	**74.5**	**64.6**	**36.7**	**19.7**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**305.0**	**332.9**	**297.3**	**282.7**	**280.8**	**287.8**	**261.7**	**238.0**	**166.1**	**59.5**		
Debt service-to-revenue and grants ratio 3/	**33.3**	**27.3**	**26.0**	**31.5**	**27.7**	**29.8**	**25.8**	**21.6**	**21.1**	**17.3**	**14.2**	**10.6**		
Gross financing need 4/	15.1	10.2	10.5	13.4	15.0	13.7	11.2	8.7	7.6	4.5	0.1	-11.1		
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	6.6	3.8	3.7	3.3	1.8	6.0	4.0	4.0	9.2	11.5	2.8	2.7	**6.3**	**8.1**
Average nominal interest rate on external debt (in percent)	1.3	1.9	2.8	2.3	1.6	1.4	1.3	1.2	1.3	1.3	2.8	2.0	**2.1**	**2.0**
Average real interest rate on domestic debt (in percent)	6.1	-1.1	-5.9	5.5	9.6	5.8	6.7	7.2	7.4	7.5	-1.3	5.7	**1.0**	**4.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	32.3	40.4	-25.0	...	...	...	...	...	...	...	...	...	**6.5**	**...**
Inflation rate (GDP deflator, in percent)	4.4	11.9	12.9	5.5	6.1	8.0	6.3	5.5	5.5	5.5	5.5	5.5	**6.2**	**5.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	-11.7	-5.1	1.1	2.5	21.1	-0.6	-2.4	-2.2	6.9	0.2	3.3	3.5	**9.9**	**3.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-19.9	-37.4	30.4	5.1	-0.1	10.4	6.1	2.5	8.6	12.1	0.2	-14.2	**-9.0**	**6.8**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus social security, central bank, government-guaranteed debt, non-guaranteed SOE debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

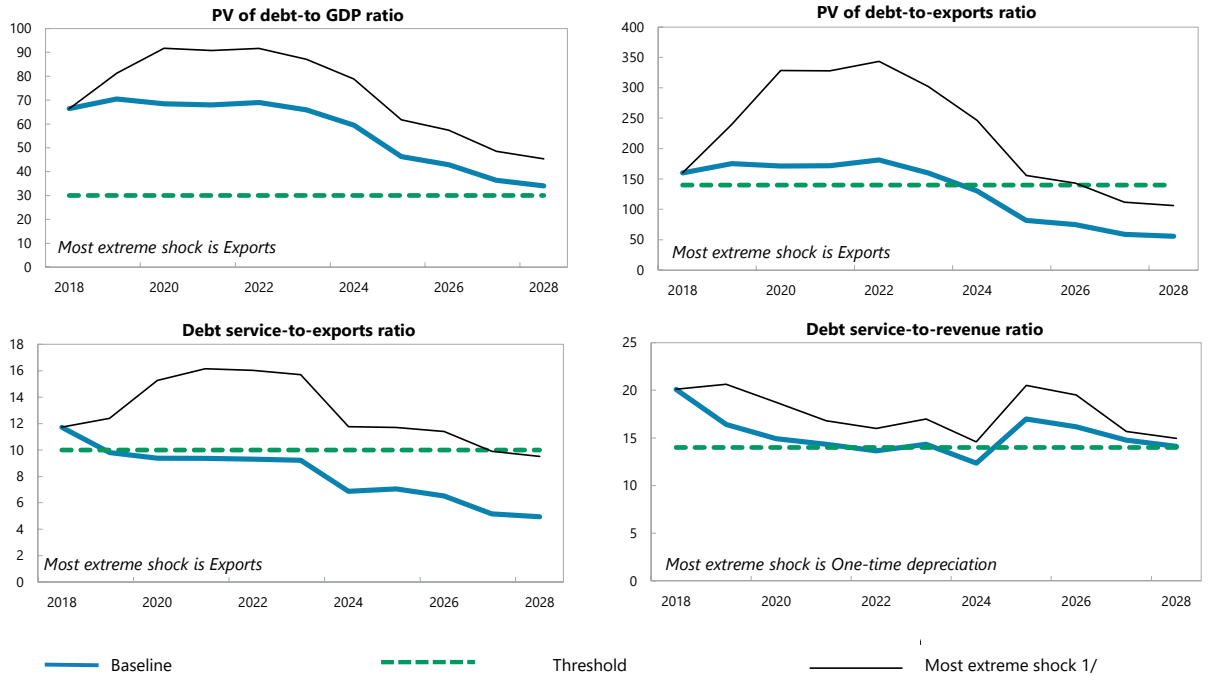
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Mozambique: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2018-28



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	No	No
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	31	24
Avg. grace period	10	6

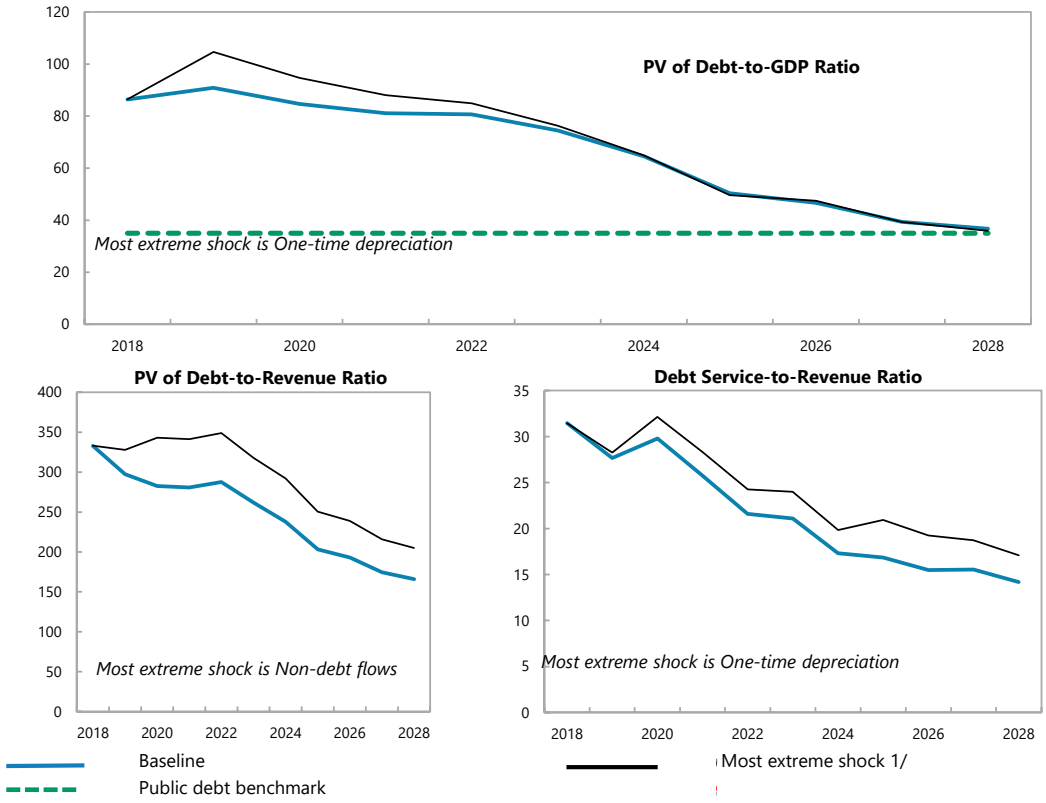
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Mozambique: Indicators of Public Debt Under Alternative Scenarios, 2018-28



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	99%	99%
Domestic medium and long-term	1%	1%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
Avg. maturity (incl. grace period)	31	24
Avg. grace period	10	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	7.6%	10.7%
Avg. maturity (incl. grace period)	4	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	3%	12.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Mozambique: Sensitivity Analysis for Key Indicators of Publicly Guaranteed External Debt, 2018-28 (in percent)

	Projections										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	**66.4**	**70.5**	**68.5**	**68.0**	**69.1**	**65.9**	**59.5**	**46.4**	**42.9**	**36.4**	**34.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	66.4	50.1	24.9	1.0	-28.5	-55.8	-72.6	-77.5	-77.4	-55.8	-39.8
B. Bound Tests											
B1. Real GDP growth	66.4	72.1	71.4	71.0	72.1	68.8	62.1	48.4	44.8	38.0	35.5
B2. Primary balance	66.4	72.2	71.7	71.1	72.1	68.8	62.1	48.5	44.9	38.1	35.8
B3. Exports	66.4	81.3	91.8	90.8	91.7	87.1	78.8	61.8	57.4	48.5	45.3
B4. Other flows 2/	66.4	79.7	86.2	85.3	86.0	81.7	73.9	58.0	53.8	45.5	42.5
B5. One-time 30 percent nominal depreciation	66.4	88.6	73.0	72.7	74.4	71.3	64.2	49.7	45.7	38.8	36.3
B6. Combination of B1-B5	66.4	83.9	83.7	82.9	83.9	79.9	72.2	56.5	52.2	44.2	41.3
C. Tailored Tests											
C1. Combined contingent liabilities	66.4	78.8	76.3	75.6	76.6	72.9	65.9	51.5	47.9	40.8	38.5
C2. Natural disaster	66.4	78.3	76.1	75.7	77.0	73.6	66.7	52.4	48.9	41.9	39.8
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	**160.1**	**175.1**	**171.5**	**171.9**	**181.2**	**159.9**	**130.3**	**81.9**	**75.1**	**58.7**	**55.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	160.1	124.5	62.5	2.6	-74.8	-135.4	-159.0	-136.8	-135.4	-90.0	-65.4
B. Bound Tests											
B1. Real GDP growth	160.1	175.1	171.5	171.9	181.2	159.9	130.3	81.9	75.1	58.7	55.8
B2. Primary balance	160.1	179.2	179.5	179.8	189.3	166.9	136.1	85.6	78.6	61.5	58.7
B3. Exports	160.1	239.3	328.5	328.1	343.6	302.0	246.6	155.8	143.4	111.8	106.3
B4. Other flows 2/	160.1	197.8	216.0	215.6	225.6	198.1	161.8	102.4	94.1	73.3	69.7
B5. One-time 30 percent nominal depreciation	160.1	175.1	145.5	146.4	155.3	137.6	111.9	69.9	63.6	49.8	47.5
B6. Combination of B1-B5	160.1	218.2	194.3	240.7	252.7	222.4	181.5	114.5	104.8	81.8	77.7
C. Tailored Tests											
C1. Combined contingent liabilities	160.1	195.7	191.1	191.2	201.0	177.0	144.4	91.0	83.8	65.9	63.1
C2. Natural disaster	160.1	198.4	194.5	195.3	206.0	182.1	149.1	94.4	87.3	69.0	66.5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	**11.7**	**9.8**	**9.4**	**9.4**	**9.3**	**9.2**	**6.9**	**7.1**	**6.5**	**5.2**	**4.9**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	11.7	9.9	8.8	7.3	5.8	5.0	2.1	5.4	2.7	-0.4	-2.4
B. Bound Tests											
B1. Real GDP growth	11.7	9.8	9.4	9.4	9.3	9.2	6.9	7.1	6.5	5.2	4.9
B2. Primary balance	11.7	9.8	9.5	9.6	9.5	9.4	7.0	7.1	6.7	5.4	5.2
B3. Exports	11.7	12.4	15.3	16.2	16.0	15.7	11.8	11.7	11.4	9.9	9.5
B4. Other flows 2/	11.7	9.8	9.9	10.4	10.4	10.1	7.6	7.5	7.5	6.5	6.3
B5. One-time 30 percent nominal depreciation	11.7	9.8	9.4	8.7	8.7	8.7	6.5	6.8	6.3	4.4	4.2
B6. Combination of B1-B5	11.7	10.9	12.2	12.3	12.2	12.0	9.0	9.0	9.0	7.2	6.9
C. Tailored Tests											
C1. Combined contingent liabilities	11.7	9.8	9.9	9.8	9.8	9.6	7.2	7.3	6.7	5.3	5.1
C2. Natural disaster	11.7	10.1	10.1	10.1	10.0	9.9	7.4	7.5	7.0	5.5	5.3
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	**20.1**	**16.4**	**14.9**	**14.3**	**13.6**	**14.3**	**12.4**	**17.0**	**16.2**	**14.8**	**14.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	20.1	16.5	13.9	11.2	8.5	7.8	3.8	13.1	6.7	-1.3	-6.9
B. Bound Tests											
B1. Real GDP growth	20.1	16.8	15.6	15.0	14.2	15.0	12.9	17.7	16.9	15.4	14.7
B2. Primary balance	20.1	16.4	15.1	14.6	13.9	14.6	12.6	17.2	16.6	15.5	14.8
B3. Exports	20.1	17.6	17.0	17.3	16.5	17.1	14.8	19.7	19.8	19.8	19.0
B4. Other flows 2/	20.1	16.4	15.8	16.0	15.2	15.7	13.6	18.1	18.7	18.6	17.9
B5. One-time 30 percent nominal depreciation	20.1	20.6	18.7	16.8	16.0	17.0	14.6	20.5	19.5	15.7	14.9
B6. Combination of B1-B5	20.1	17.5	16.9	16.4	15.6	16.2	14.0	18.9	19.4	18.0	17.3
C. Tailored Tests											
C1. Combined contingent liabilities	20.1	16.4	15.7	15.0	14.3	14.9	12.9	17.5	16.6	15.2	14.5
C2. Natural disaster	20.1	16.4	15.5	14.9	14.2	14.9	12.9	17.5	16.6	15.2	14.6
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Includes official and private transfers and FDI.

Table 4. Mozambique: Sensitivity Analysis for Key Indicators of Public Debt, 2018-28

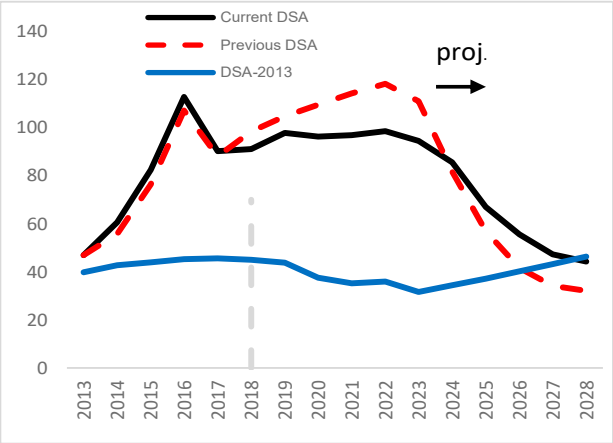
	Projections										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	**86.4**	**90.9**	**84.7**	**81.1**	**80.6**	**74.5**	**64.6**	**50.3**	**46.7**	**39.4**	**36.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	86	86	82	77	77	74	71	71	72	72	71
B. Bound Tests											
B1. Real GDP growth	86	93	89	86	87	81	71	56	53	45	43
B2. Primary balance	86	93	88	84	84	77	67	52	49	41	38
B3. Exports	86	97	100	96	95	88	77	60	56	47	44
B4. Other flows 2/	86	100	103	99	98	90	79	62	58	49	45
B5. One-time 30 percent nominal depreciation	86	105	95	88	85	76	65	50	47	39	36
B6. Combination of B1-B5	86	89	84	78	78	72	62	48	45	38	35
C. Tailored Tests											
C1. Combined contingent liabilities	86	99	93	89	88	81	71	56	52	44	41
C2. Natural disaster	86	99	93	89	89	82	72	56	53	45	43
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	**332.9**	**297.3**	**282.7**	**280.8**	**287.8**	**261.7**	**238.0**	**203.1**	**193.1**	**174.7**	**166.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	333	284	274	270	275	263	261	286	295	316	320
B. Bound Tests											
B1. Real GDP growth	333	303	296	298	308	283	260	225	218	201	194
B2. Primary balance	333	303	293	292	299	272	248	212	202	183	174
B3. Exports	333	316	333	332	339	309	283	243	232	209	199
B4. Other flows 2/	333	328	343	342	349	318	292	251	239	216	205
B5. One-time 30 percent nominal depreciation	333	348	320	307	305	269	241	201	197	175	163
B6. Combination of B1-B5	333	294	281	271	278	252	229	195	185	167	159
C. Tailored Tests											
C1. Combined contingent liabilities	333	325	309	307	315	286	262	224	214	195	186
C2. Natural disaster	333	323	308	308	316	289	265	228	218	200	192
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	**31.5**	**27.7**	**29.8**	**25.8**	**21.6**	**21.1**	**17.3**	**16.8**	**15.5**	**15.5**	**14.2**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	31	27	29	25	20	20	18	20	19	21	19
B. Bound Tests											
B1. Real GDP growth	31	28	31	27	23	22	18	18	17	17	16
B2. Primary balance	31	28	30	26	22	21	18	17	16	16	15
B3. Exports	31	28	30	27	23	22	18	18	17	19	17
B4. Other flows 2/	31	28	31	27	23	22	19	18	18	19	18
B5. One-time 30 percent nominal depreciation	31	28	32	28	24	24	20	21	19	19	17
B6. Combination of B1-B5	31	27	29	25	21	21	17	16	15	15	14
C. Tailored Tests											
C1. Combined contingent liabilities	31	28	31	26	23	22	18	17	16	16	15
C2. Natural disaster	31	28	31	27	23	22	18	18	16	16	15
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

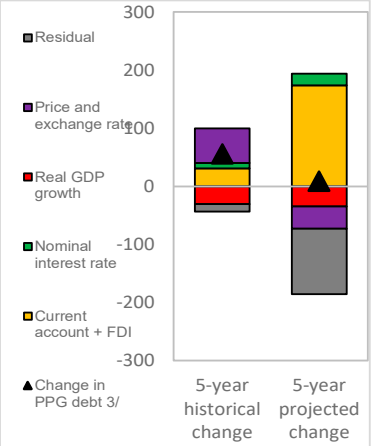
Figure 3. Mozambique: Drivers of Debt Dynamics - Baseline Scenario

External debt

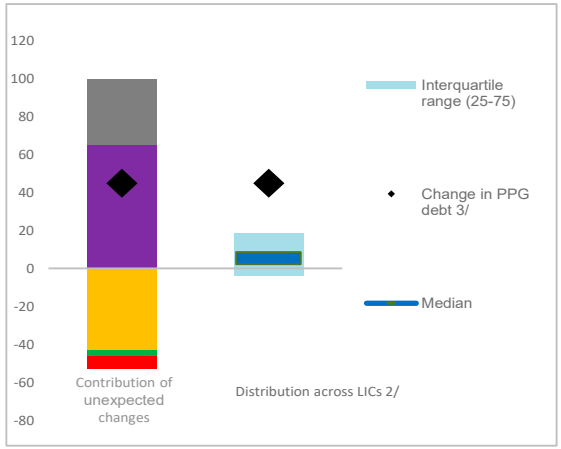
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

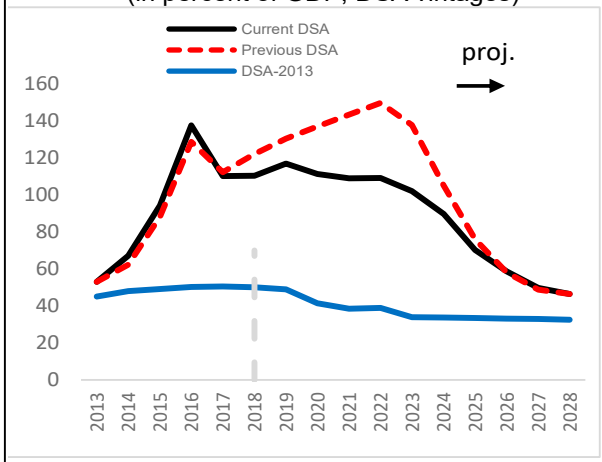


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

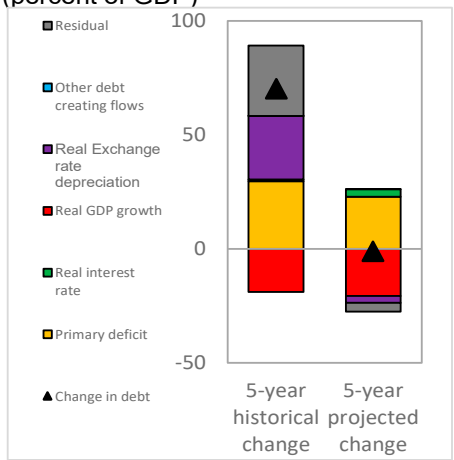


Public debt

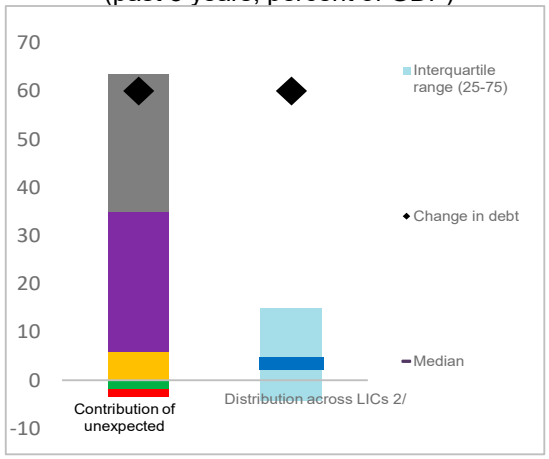
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



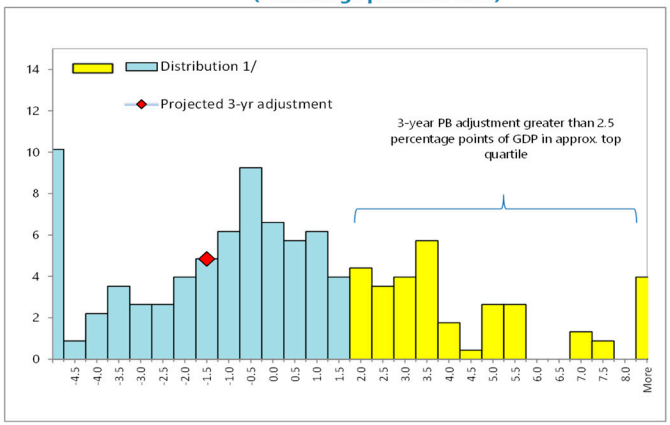
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

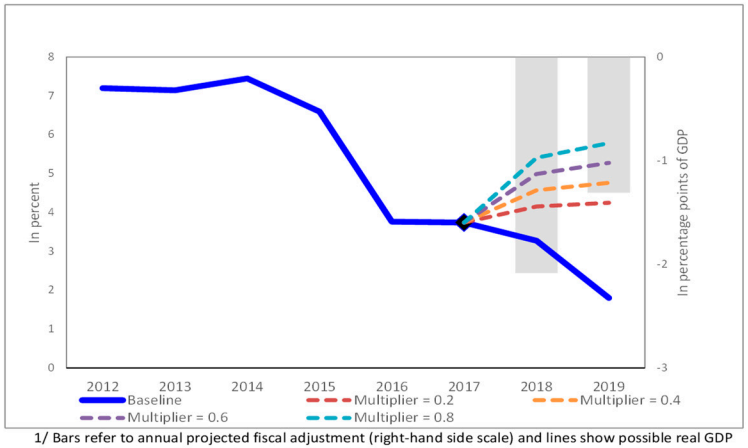
Figure 4. Mozambique: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



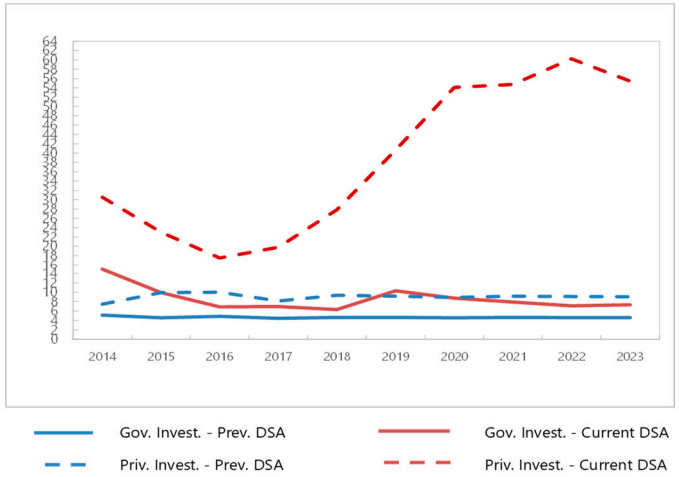
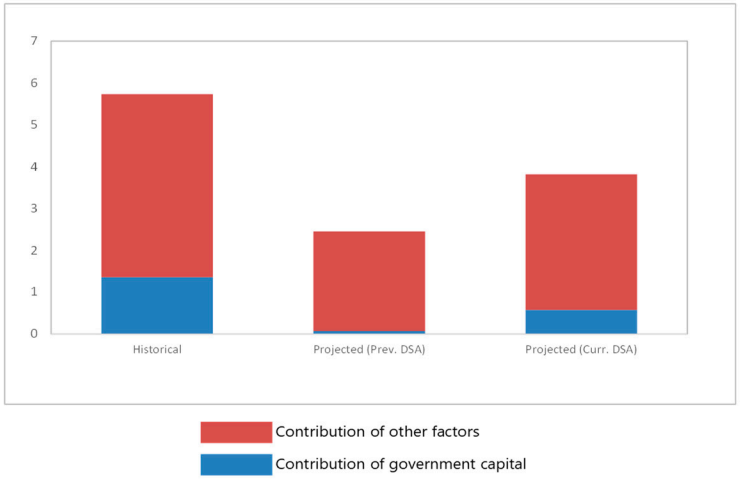
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)